

**38** Penicillin is an antibiotic used to treat infections caused by bacteria.

Which events explain how bacteria become resistant to penicillin?

- 1 The number of bacteria with the allele for resistance to penicillin increases.
- 2 Resistance to penicillin is the result of a mutation in a bacterium.
- 3 Bacteria with the allele for resistance to penicillin pass the allele to their daughter cells.
- 4 The mutation for resistance to penicillin is always caused by the presence of penicillin.

- A** 1, 2, 3 and 4
- B** 1, 2 and 3 only
- C** 1 and 4 only
- D** 2 and 3 only